

UNIVERSITI TEKNOLOGI PETRONAS

Project Report 2

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BACHELOR OF COMPUTER SCIENCE

MODELLING AND SIMULATION

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1. Introduction

In our first submission, we used a Python agent-based model to show that a nanobot swarm following chemical signals heals a skin wound much faster than a random swarm or no nanobots. For this submission, we rebuilt that model as a cellular automaton (CA) in NetLogo [1] to check if the result still holds. A CA is a grid whereby every cell has one of a few states and all cells update at the same time using only their neighbours [2]. The code is at https://github.com/Nowkee/M8S_Assignment under Second-Submission.

2. The Model

The skin is a 40 by 40 grid of patches with an oval wound of 331 cells, and there are no turtles at all. Each cell is healthy, damaged, infected or healing, and starts with four random infection patches that grow and spread. Unhealed cells give off a chemical signal, while the body slowly heals from the edges but never an infected cell. We kept every number from the Python model.

Instead of agents, each cell also holds at most one nanobot, with 20 scouts, 40 cleaners and 40 carriers starting in a gel ring around the wound. A bot moves by picking the free neighbour with the strongest signal (randomly 15% of the time), and each empty cell pulls in one bot that picked it. All cells work out their next values first, then switch at the same time.

Healing time is the number of steps until 95% of the wound is healthy. Each case was run 20 times for 400 steps using BehaviorSpace.

3. Results

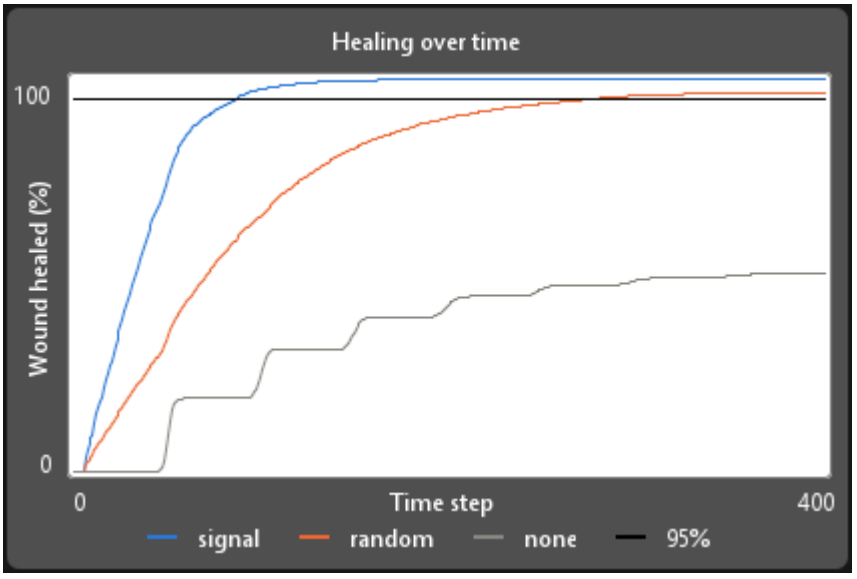


Figure 1 shows the average percentage of the wound healed over time in NetLogo, with the 95% finish line.

Case	Runs that reached 95% (out of 20)	Mean healing time (steps)	Healed after 400 steps	Infected cells after 400 steps
Signal-following swarm	20	86.9 (sd 4.9)	100%	0
Random swarm	15	243.9 (sd 48.7)	96.6%	11.1
No nanobots	0	Did not heal	50.5%	162.8

Table 1 shows the results from 20 runs of each case. The random swarm time only averages its 15 finished runs.

The signal-following swarm healed in every run and was about 2.8 times faster than the random swarm, whose bots drift into healthy skin and waste their fuel. Without nanobots, healing got stuck at around 50% as the infection grew to about 163 cells.



Figure 2 shows the same wound (seed 3) under each case at steps 0, 30, 60 and 150.

4. Discussion and Conclusion

Compared to our Python model (72.5 steps), the signal-following swarm is about 14 steps slower in the CA, since only one bot can sit on a cell, so carriers can no longer pile up on the same square. However, the parameters were picked by us, so the step counts are not real time, and the model is flat and ignores blood flow. All in all, even with only local rules and synchronous updates, the way the swarm moves still matters as much as having the nanobots there.

References

[1] U. Wilensky, NetLogo. Center for Connected Learning and Computer-Based Modeling, Northwestern University, 1999. [Online]. Available: https://ccl.northwestern.edu/netlogo/

[2] J. G. Freire and C. C. DaCamara, "Using cellular automata to simulate wildfire propagation and to assist in fire management," Nat. Hazards Earth Syst. Sci., vol. 19, no. 1, pp. 169-179, 2019, doi: 10.5194/nhess-19-169-2019.

